ENGSCI 205 – Special Topic

Section A: Data Science
NEUQ10 – Regression

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Adapted from material made by Andreas W. Kempa-Liehr

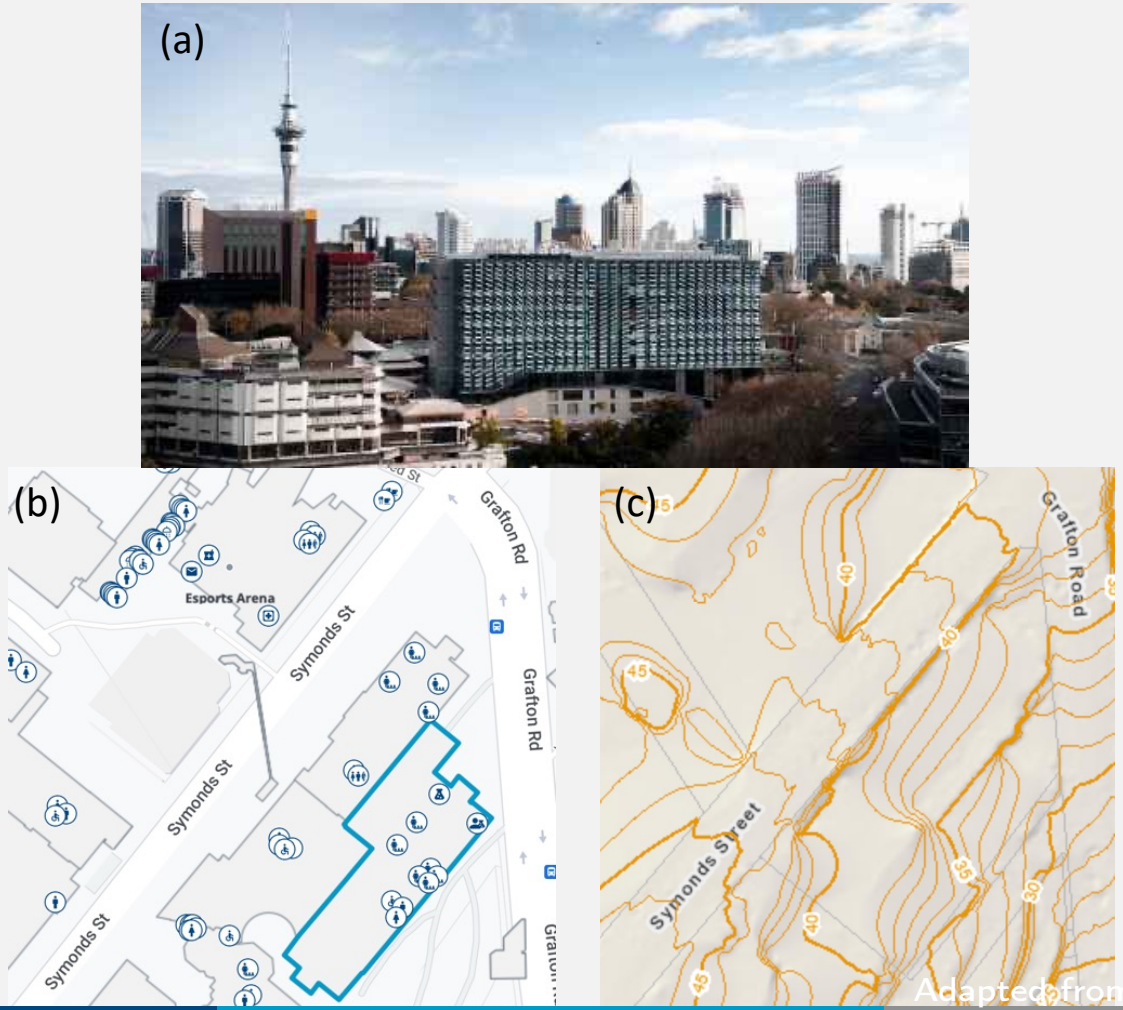
Outline

1. Models
2. Learning from data
3. Machine Learning explained
4. Model-based vs. instance based learning
5. Linear Regression from an optimization perspective
6. Main challenges of machine learning

What is a model?

A *model* is a simplified representation of our world with a very specific use case.

Example (Corner of Symonds St and Grafton Rd)



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Model classes in artificial intelligence

- *Learning model* or *learning algorithm* or *machine learning model*
  - Is always problem specific
  - Includes the preprocessing of input/observed/collected/sampled data
  - Approximates a mathematical function from provided data by optimising a performance measure
  - Has hyperparameters parameterising the learning process
- *Learned model* or *fitted model* or *machine learning model*
  - Depends on data provided during learning
  - Can be applied to new data/observations
  - Works well only for data, which are *similar* to the training data
  - Degrades over time
- *Decision model*
  - Converts the output of learned (machine learning) models into actions
  - Requires probabilistic predictions in order to minimise costs

How do you learn?

Learning is basically *building a mental model*.

The data on the right hand side have been collected at a fruit store.
What can you tell me about the data in this table?

mass	width	height	color_score
192	8.4	7.3	0.55
180	8.0	6.8	0.59
176	7.4	7.2	0.60
86	6.2	4.7	0.80
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178	7.1	7.8	0.92
172	7.4	7.0	0.89
166	6.9	7.3	0.93

Classification: Labels are given in column `fruit_name`

In *supervised learning*, the training data you feed to the algorithm includes the desired solutions, called *labels*.

How would you discriminate apples and mandarins?

Classification Model

fruit_name	fruit_subtype	mass	width	height	color_score
apple	granny_smith	192	8.4	7.3	0.55
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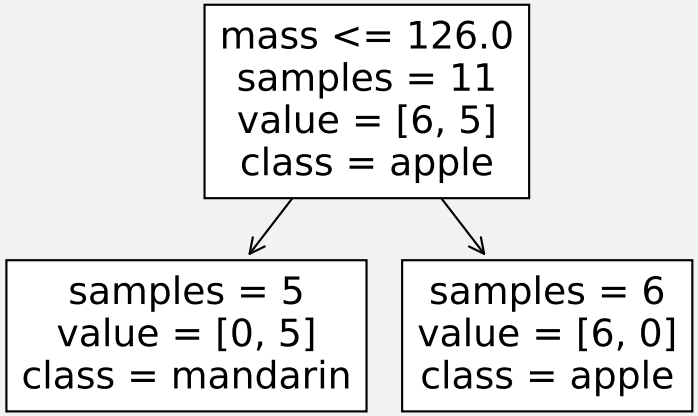

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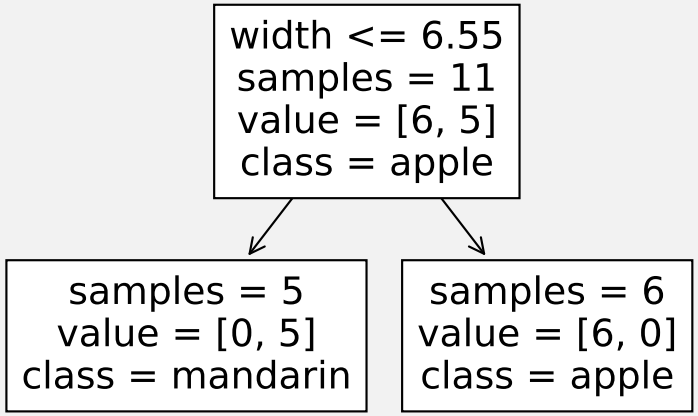

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How would you estimate the fruit mass?

Feature (Column) Selection

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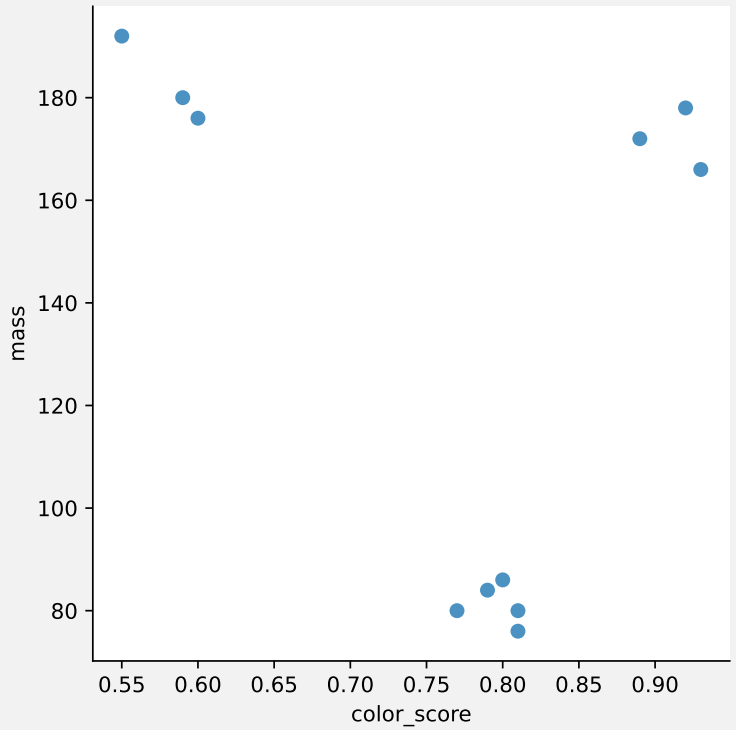

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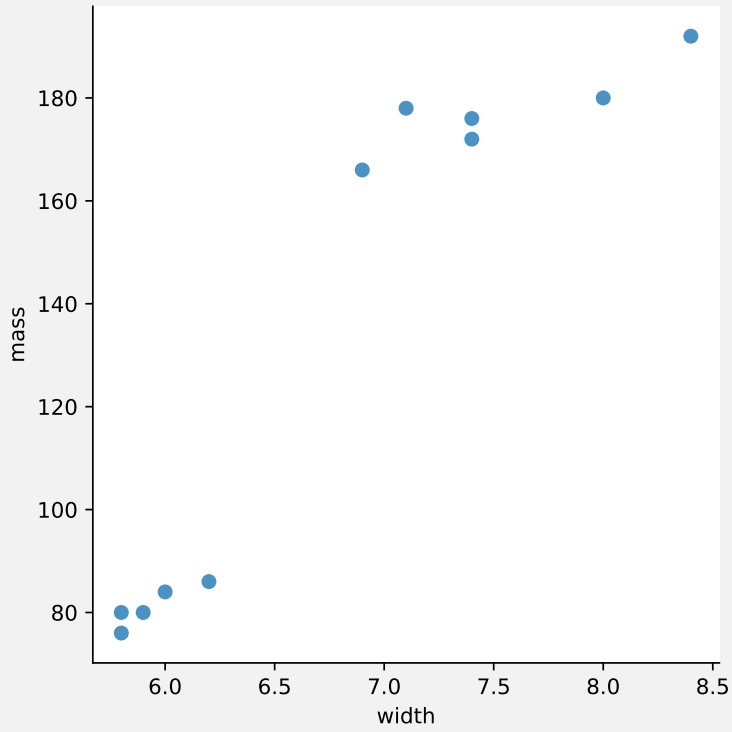

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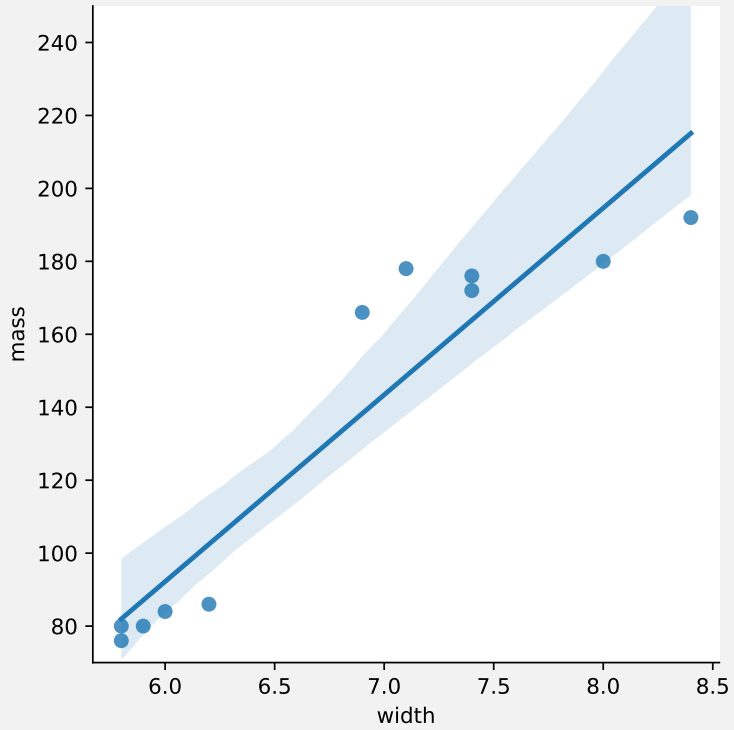

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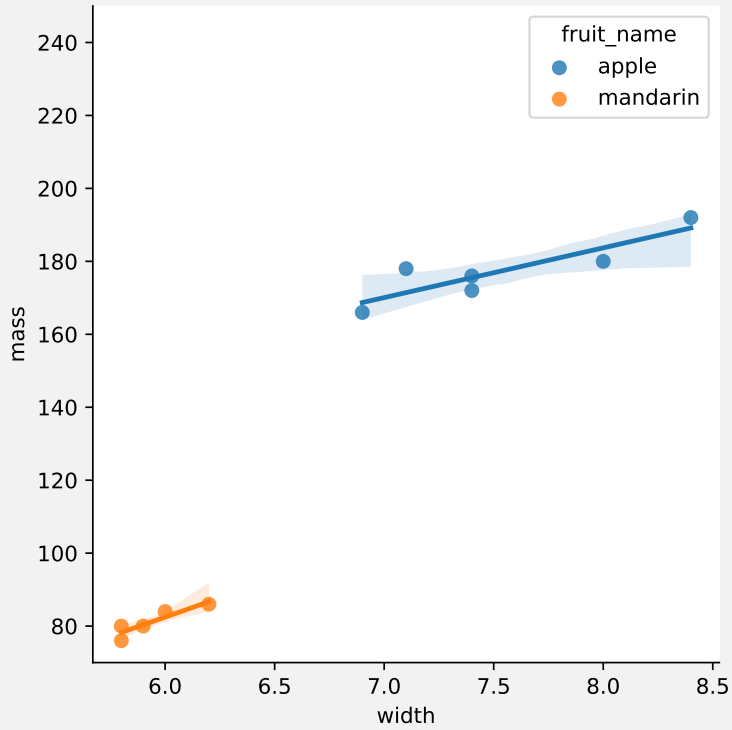

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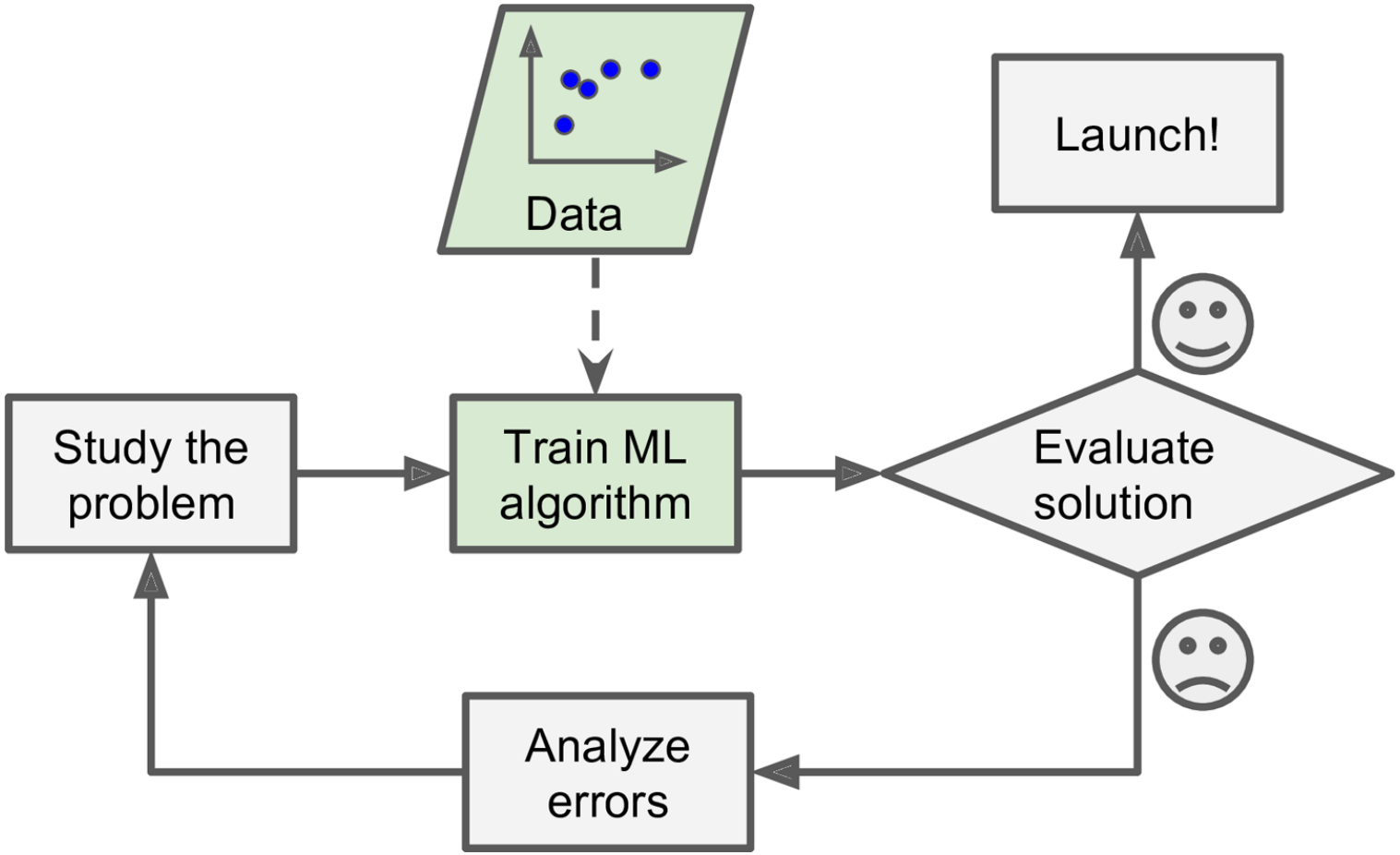
Machine learning

Training a ML algorithm is also called *fitting*.



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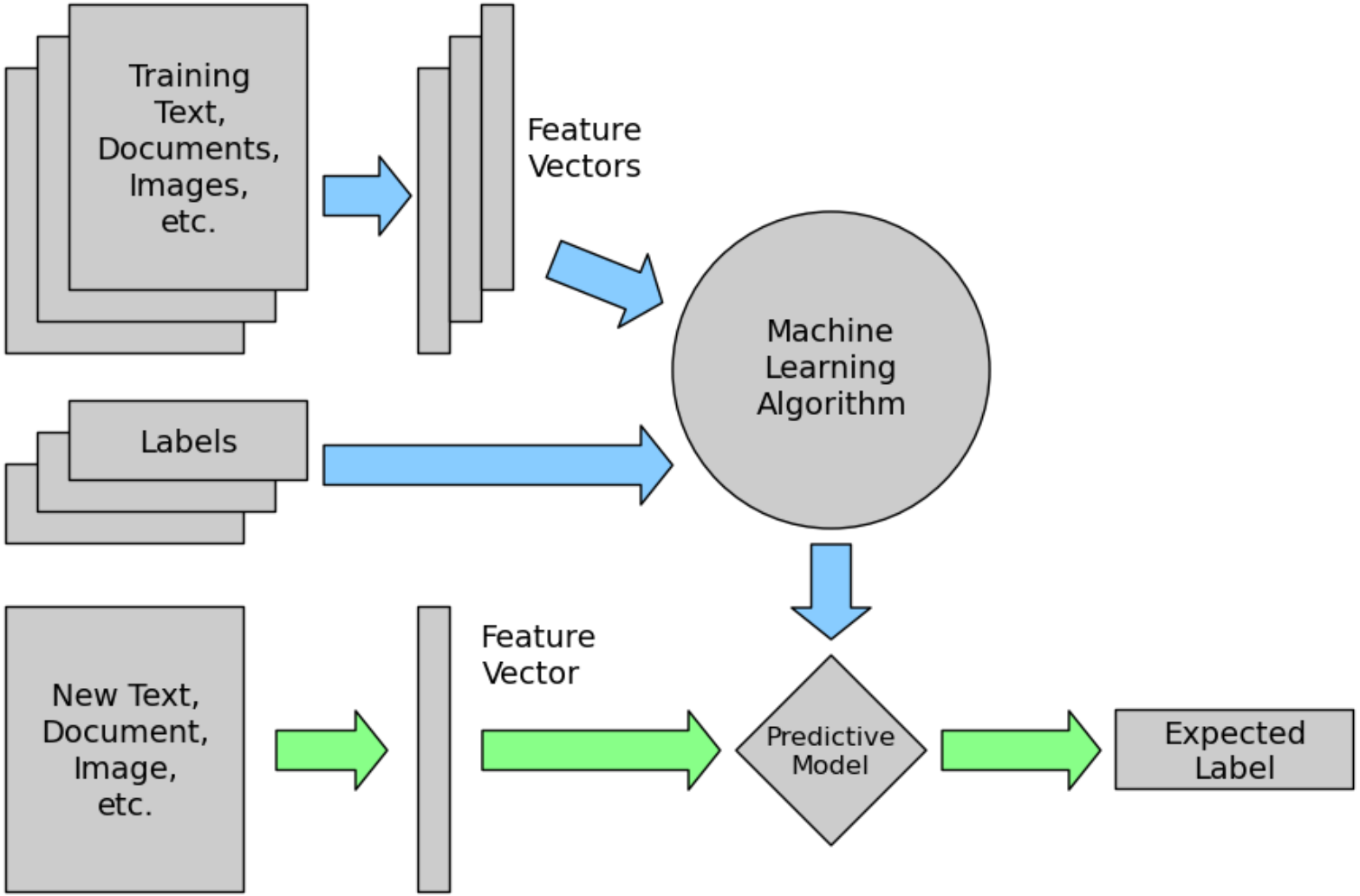
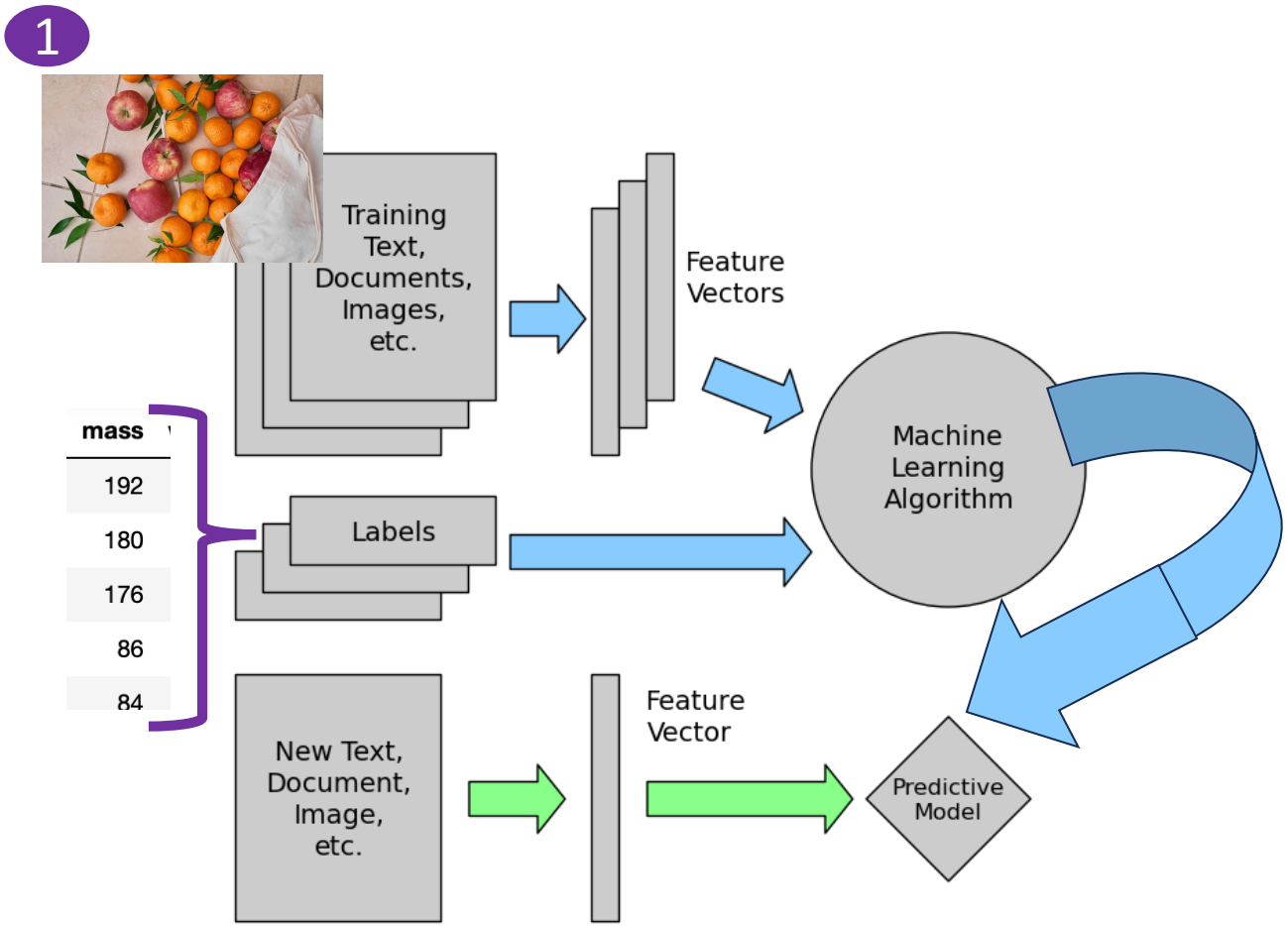
Flowchart of supervised learning

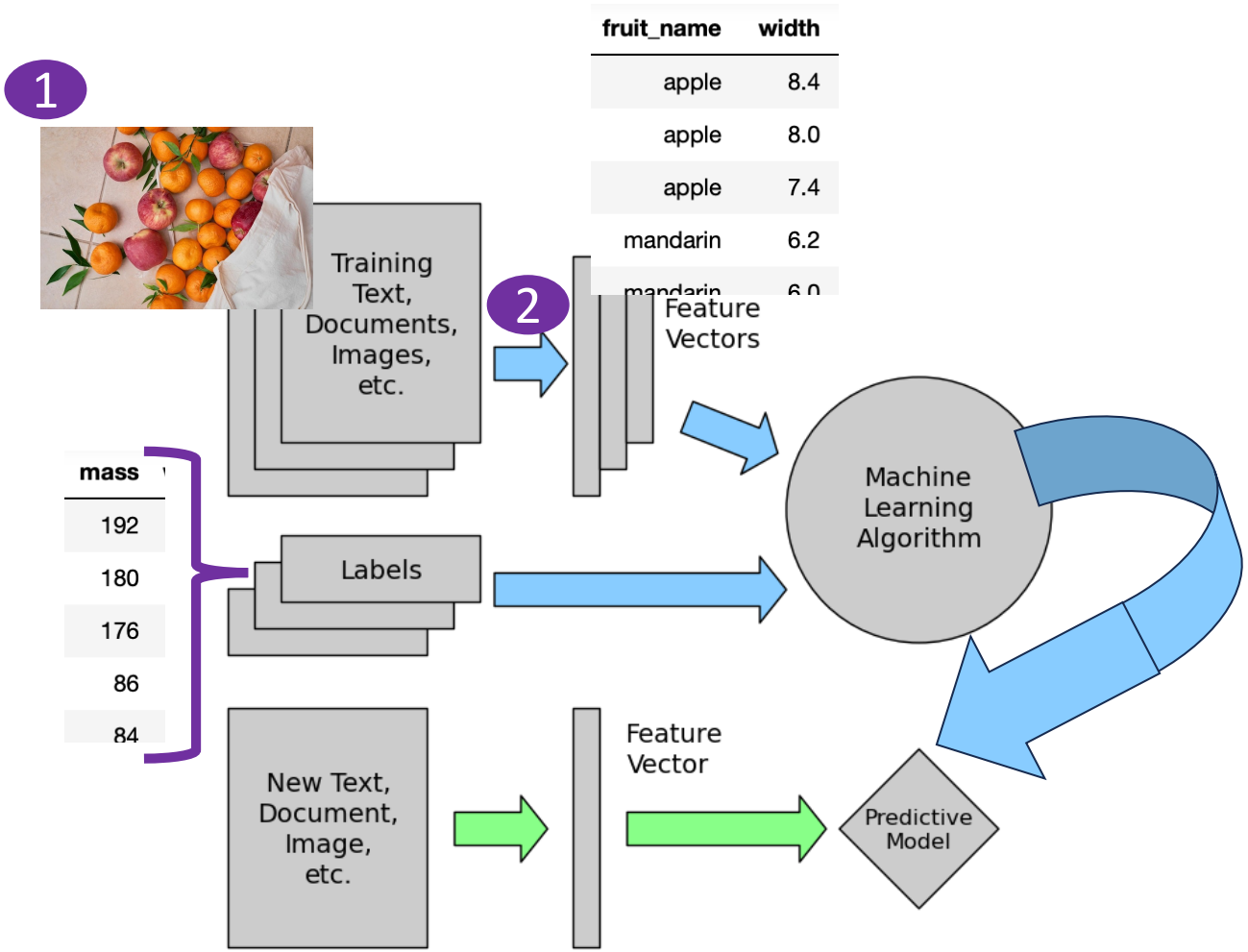


VanderPlas2012_sklearn

Machine learning in action - (1) Data collection



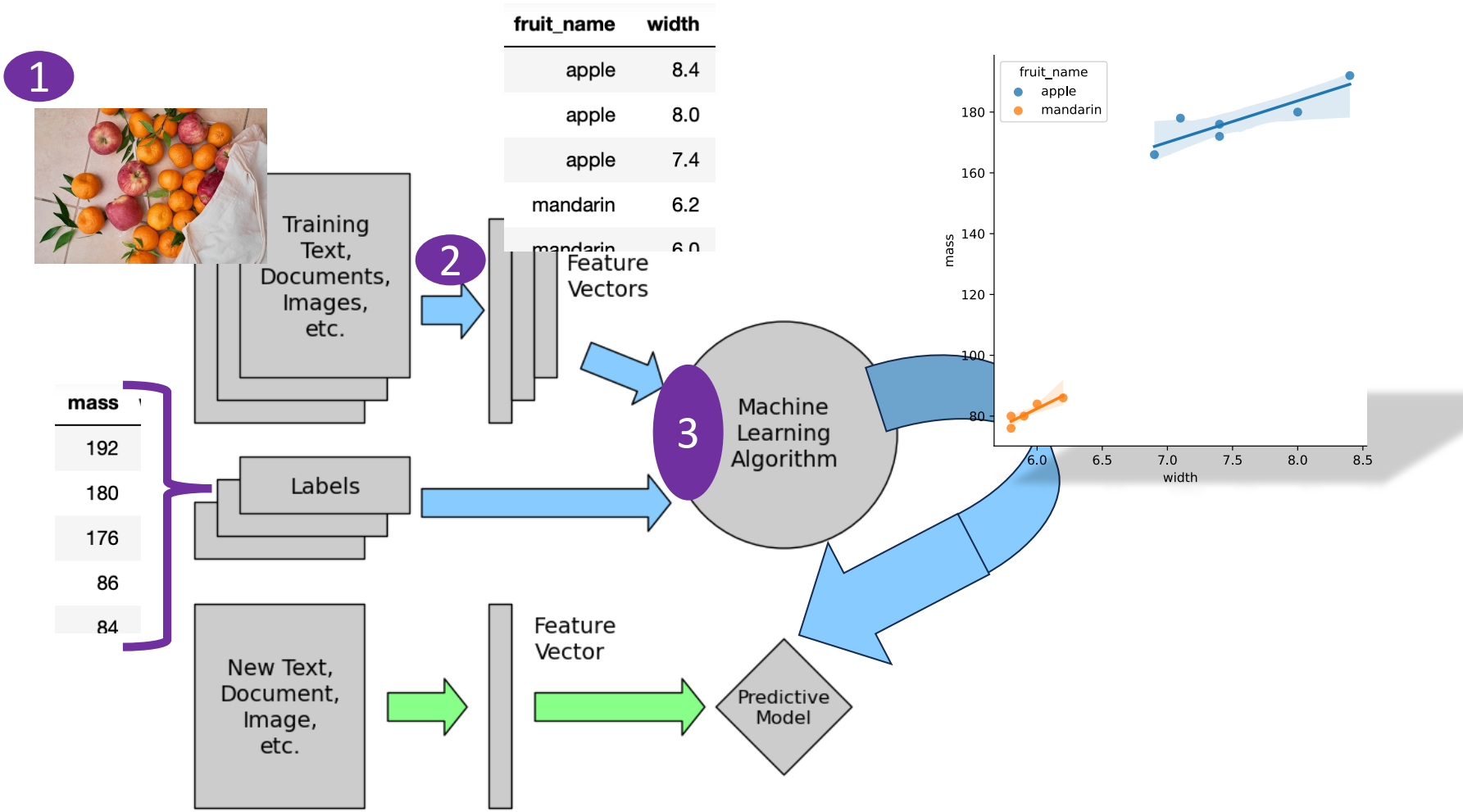
Machine learning in action - (2) Feature extraction



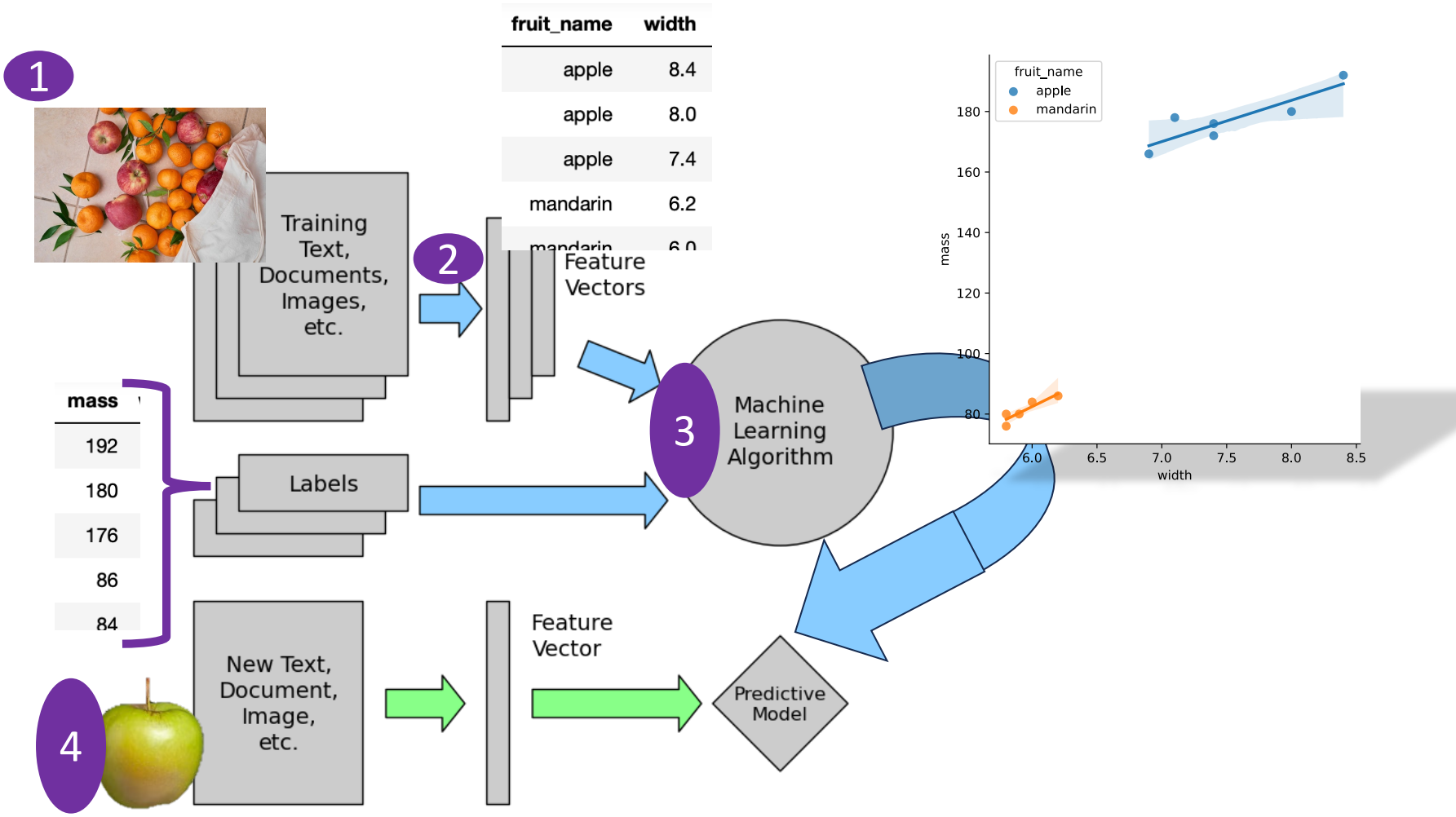
fruit_name	width
apple	8.4
apple	8.0
apple	7.4
mandarin	6.2
mandarin	6.0

mass
192
180
176
86
84

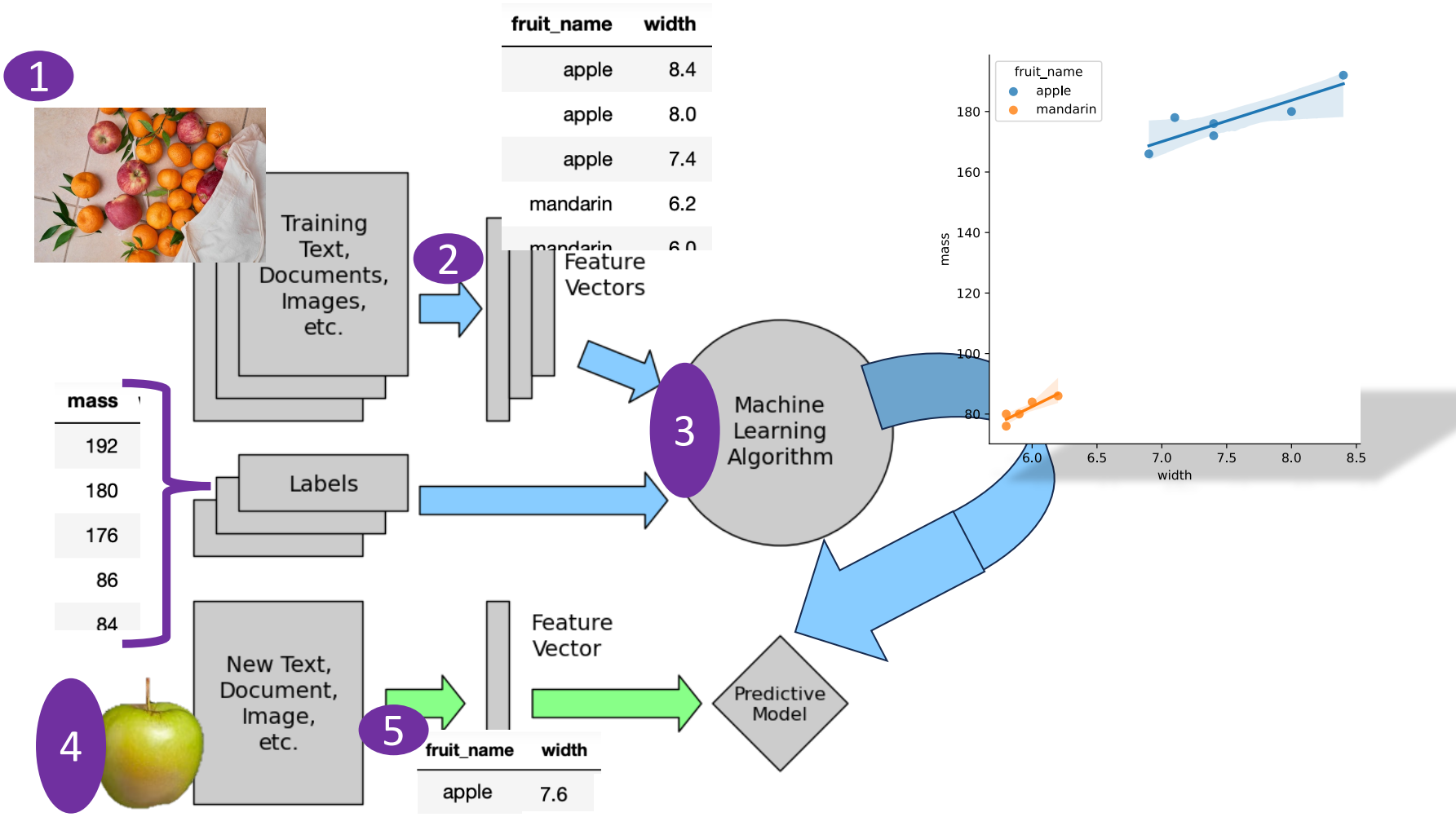
Machine learning in action - (3) Model fitting/training



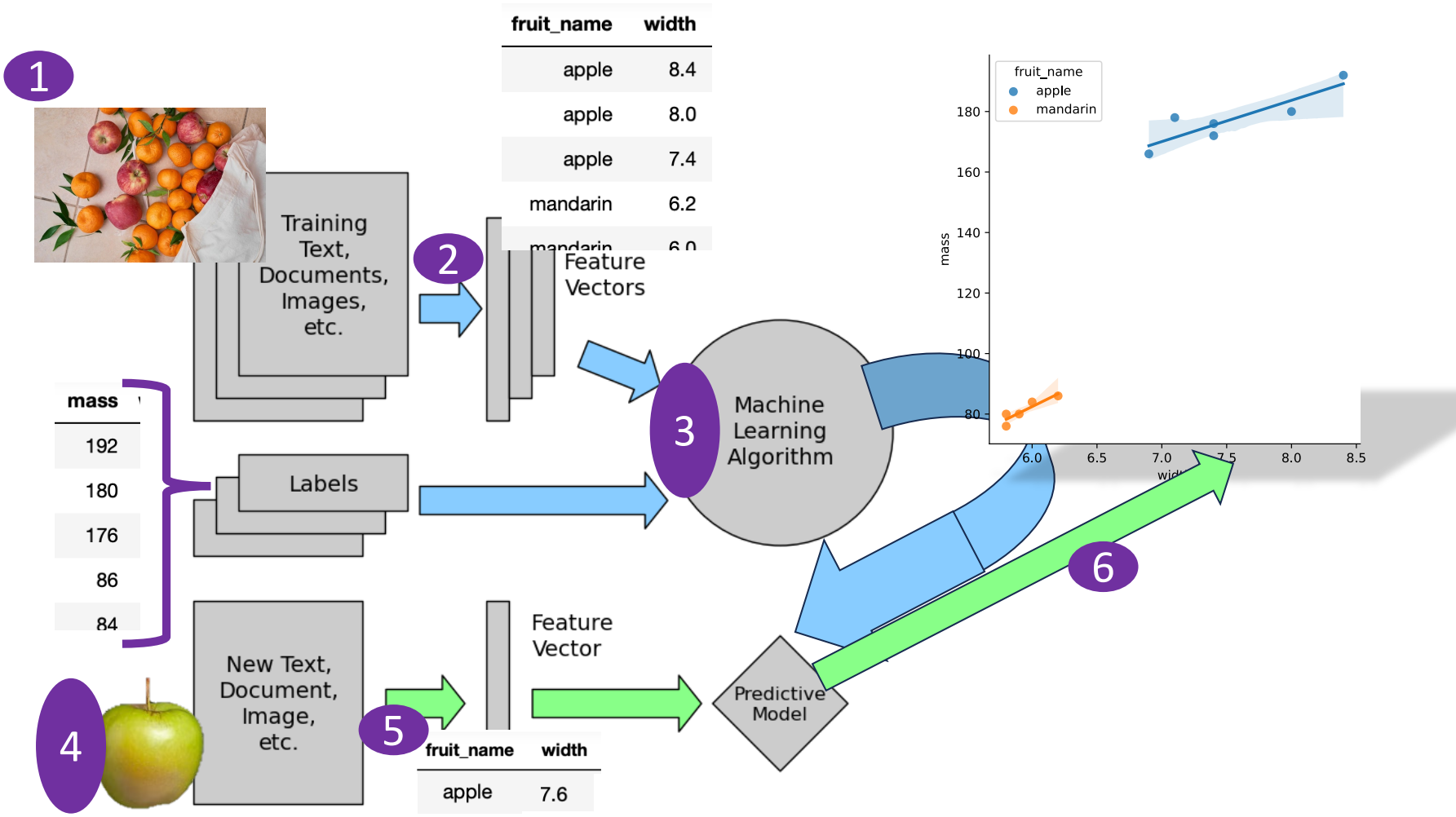
Machine learning in action - (4) New observation



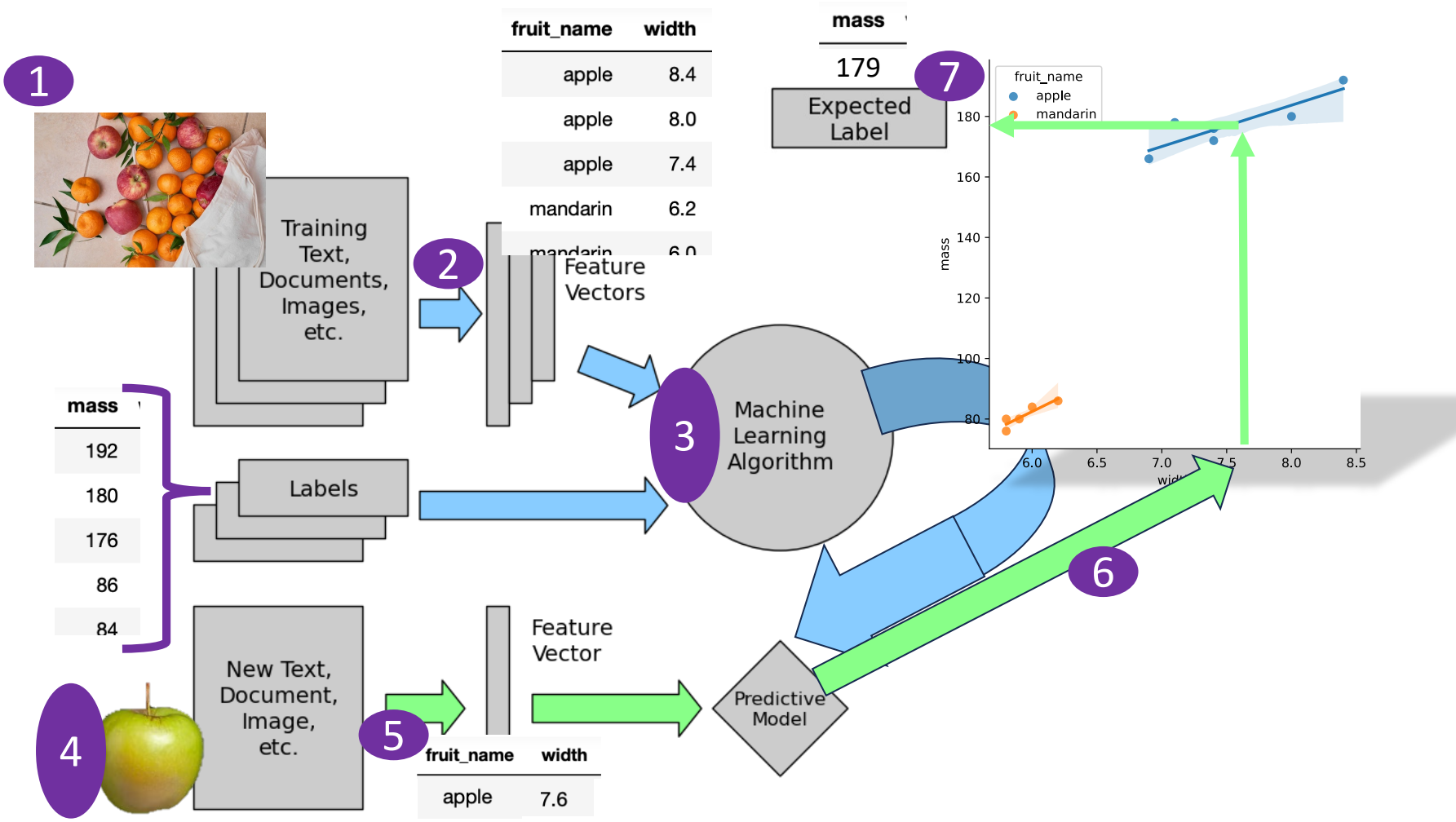
Machine learning in action - (5) Feature extraction



Machine learning in action - (6) Use model



Machine learning in action - (7) Prediction



Supervised learning

Informal Definition

In *supervised learning*, the training data you feed to the algorithm includes the desired solutions, called *labels*.

Formal Definition

In the predictive or supervised learning approach, the goal is to learn a mapping $\hat{M}$ from inputs $\vec{x} \in \mathbb{R}^D$ to outputs $y \in \mathcal{Y}$, given a labeled set of input-output pairs $\mathcal{D} = \{(\vec{x}_i, y_i)\}_{i=1}^N$ in order to predict y from a given $\vec{x}$:

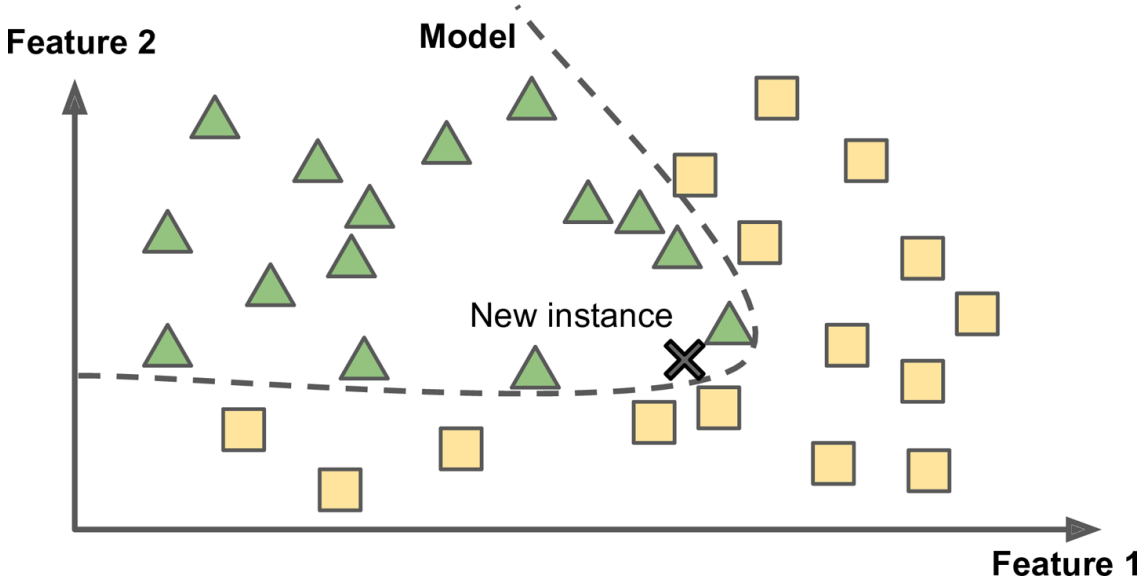
$$y = \hat{M}_{\vec{\alpha}}(\vec{x}|\mathcal{D}, \vec{\theta}) + \epsilon.$$

Here $\mathcal{D}$ is called the training set, N is the number of training examples, $\vec{\theta}$ denotes learned coefficients of model, $\vec{\alpha}$ denotes hyperparameters of the learning algorithm, and ϵ is the approximation error.

Model-based learning

Informal definition

Model-based learning systems generalize from a set of examples by building a mathematical model of these examples, then use that model to make predictions.



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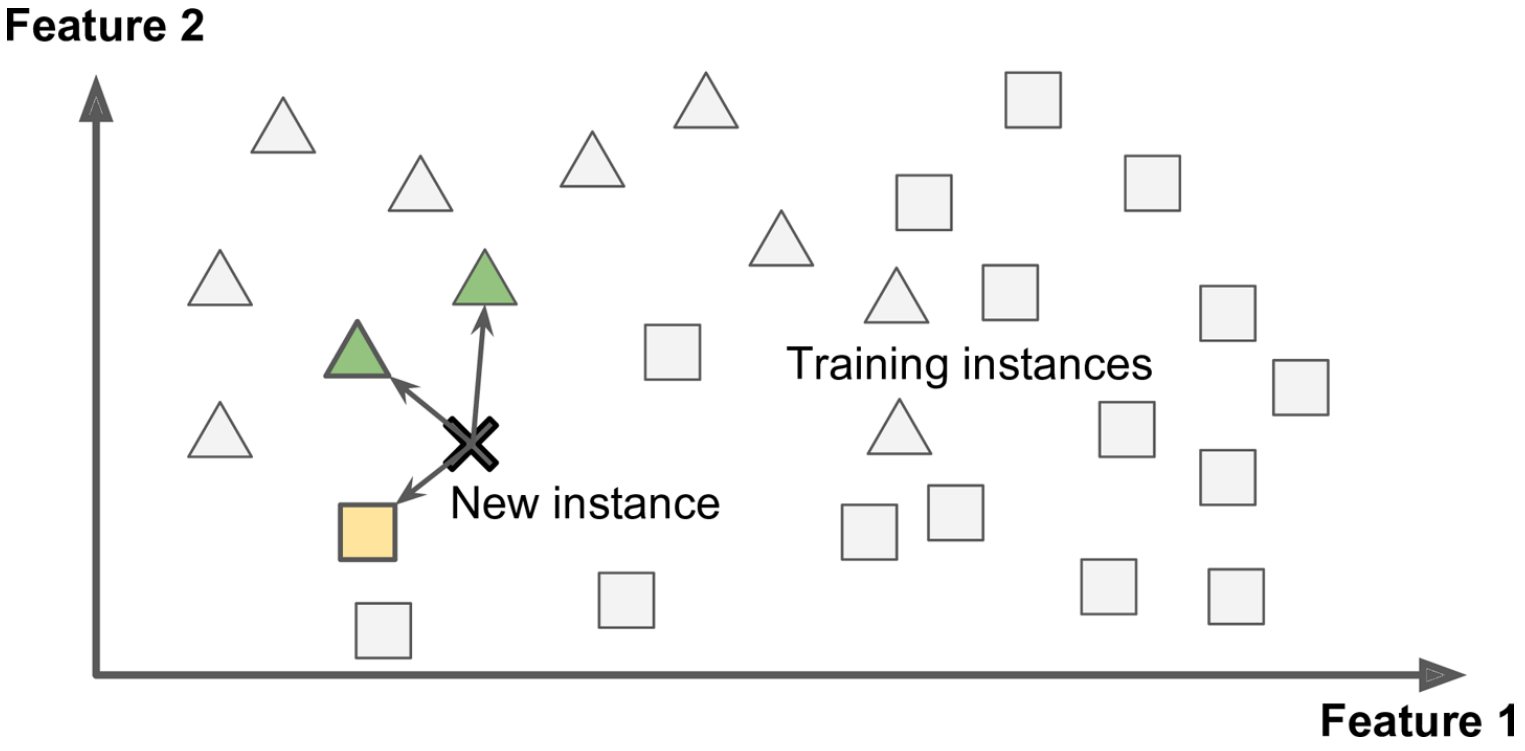
Instance-based learning (e.g. Nearest Neighbors)

Definition

Instance-based learning systems learn the examples by heart, then generalizes to new cases using a similarity measure.



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Model-based learning (Definitions)

Formal defintion

Given a set of training examples $\mathcal{D} = \{(\vec{x}_i, y_i)\}_{i=1}^N$, a model-based learning system estimates a function $\hat{M}_{\vec{\alpha}}(\vec{x})$, which minimizes a problem specific loss $L(y, M_{\vec{\alpha}}(\vec{x}))$:

$$\hat{M}_{\vec{\alpha}}(\vec{x}) = \arg\min_{M_{\vec{\alpha}}} \sum_{i=1}^{N} \mathcal{L}(y_i, M_{\vec{\alpha}}(\vec{x}_i)) =$$

$$M_{\vec{\alpha}}(\vec{x}|\hat{\vec{\theta}}) = \arg\min_{\vec{\theta}} \sum_{i=1}^{N} \mathcal{L}(y_i, M_{\vec{\alpha}}(\vec{x}_i|\vec{\theta}))$$

Squared loss

$\mathcal{L}(y, M(\vec{x})) = (y - M(\vec{x}))^2$
minimizes Mean Squared Error

Absolute loss

$\mathcal{L}(y, M(\vec{x})) = |y - M(\vec{x})|$
minimizes Mean Absolute Error

Performance measures for regression problems (1/2)

The performance measure P needs to be selected at the very beginning of every data science / machine learning project.

Mean Squared Error (MSE)

$$\text{MSE}_{\mathcal{D}} = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{M}_{\vec{\alpha}}(\vec{x}_i) \right)^2$$

- The unit of MSE is the unit of y^2. This is hard to interpret!
- The MSE depends on the data set $\mathcal{D} = \{(\vec{x}_i, y_i)\}_{i=1}^{N}$.
- MSE is very sensitive to outliers.
- Variants of MSE are the Root Mean Square Error $\text{RMSE}_{\mathcal{D}}$ and R^2.
- MSE is used for computing the deterministic solution of linear regression problems.

Performance measures for regression problems (2/2)

The performance measure P needs to be selected at the very beginning of every data science / machine learning project.

Mean absolute error (MAE)

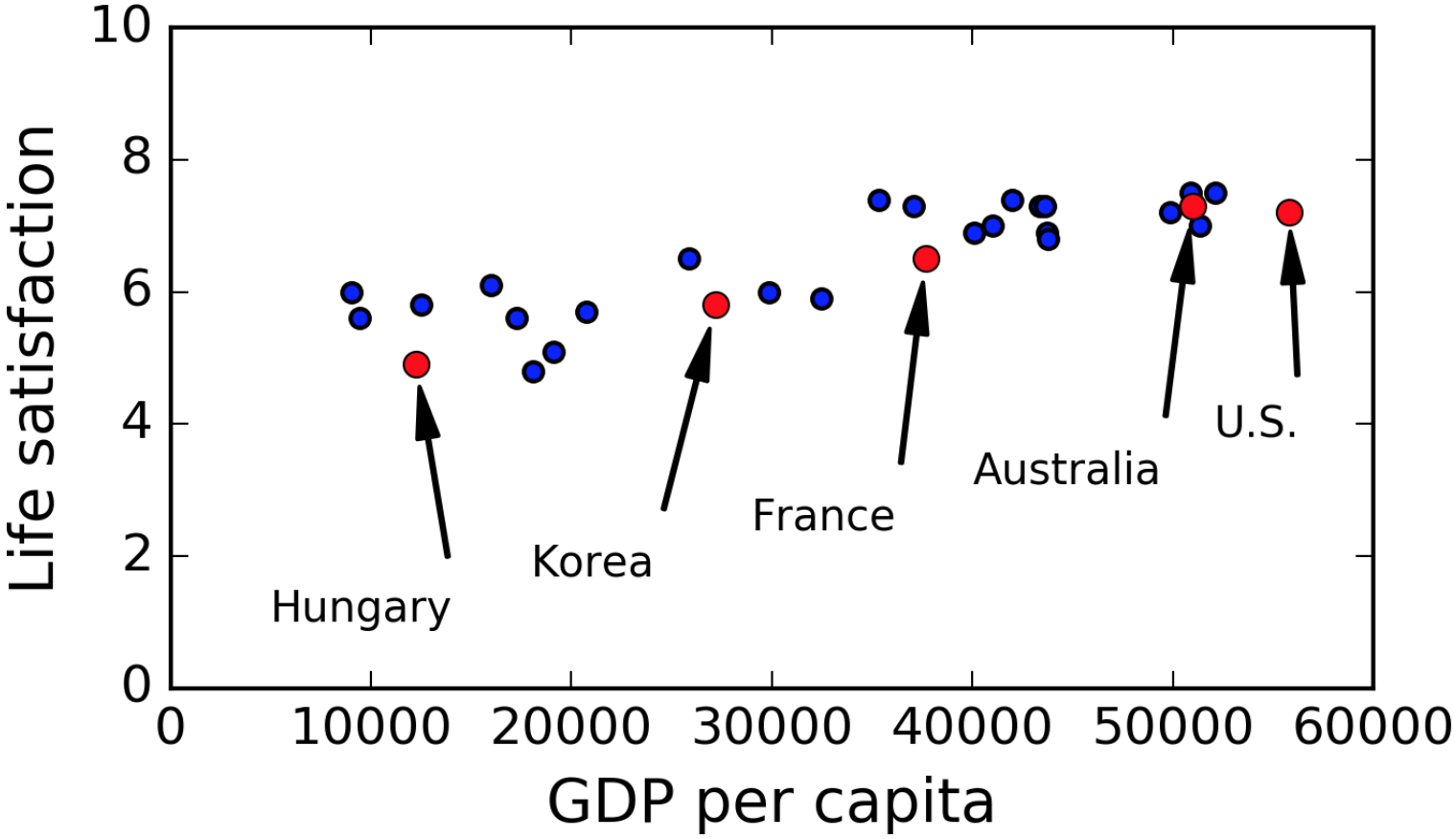
$$\text{MAE}_{\mathcal{D}} = \frac{1}{N} \sum_{i=1}^{N} |y_i - M_{\vec{\alpha}}(\vec{x}_i)|$$

- The unit of MAE is the unit of y.
- The MSE depends on the data set $\mathcal{D} = \{(\vec{x}_i, y_i)\}_{i=1}^{N}$.
- MAE is more robust to outliers than MSE.
- MAE is an important performance measure for replenishment problems.
- MAE is typically minimized by quantile regression or ensemble based algorithms (e.g. Random Forest).

Adapted from material made by And

Model-based learning: Life satisfaction/GDP example

Joining *Better Life Index data* from the OECD's website as well as stats about *GDP* per capita from the IMF's website



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Machine learning nomenclature

Let y_i be the observations of `life_satisfaction`, and x_i be `GDP_per_capita`:

Linear regression model in statistics

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$

Machine learning perspective

$\hat{y}_i = w_0 + w_1 x_i$

- $\hat{y}_i$ is the estimated value of y_i given x_i.
- w_0 and w_1 are called *weights*.
- weak exogeneity (no observation errors)
- Most machine learning algorithms do not make any assumptions about ϵ_i and just care to optimize the performance.

In-sample estimation of w_0 and w_1

$$MSE_{\mathcal{D}}(w_0, w_1) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{w}_0 - \hat{w}_1 x_i)^2$$

$$\frac{\partial}{\partial \hat{w}_0} MSE_{\mathcal{D}}(\hat{w}_0, \hat{w}_1) = \frac{2}{N}\sum_{i=1}^{N}(-1)\,(y_i - \hat{w}_0 - \hat{w}_1 x_i) = 0$$

$$\frac{\partial}{\partial \hat{w}_1} MSE_{\mathcal{D}}(\hat{w}_0, \hat{w}_1) = \frac{2}{N}\sum_{i=1}^{N}(-x_i)\,(y_i - \hat{w}_0 - \hat{w}_1 x_i) = 0$$

Analytical solution

$$\hat{w}_0 \sum_{i=1}^{N} 1 + \hat{w}_1 \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} y_i$$

$$\hat{w}_0 \sum_{i=1}^{N} x_i + \hat{w}_1 \sum_{i=1}^{N} x_i^2 = \sum_{i=1}^{N} y_i x_i$$

$$\begin{pmatrix} N & \sum_{i=1}^{N} x_i \\ \sum_{i=1}^{N} x_i & \sum_{i=1}^{N} x_i^2 \end{pmatrix} \begin{pmatrix} \hat{w}_0 \\ \hat{w}_1 \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^{N} y_i \\ \sum_{i=1}^{N} y_i x_i \end{pmatrix} \Rightarrow \mathbf{X}^\mathsf{T}\mathbf{X}\hat{\vec{\theta}} = \mathbf{X}^\mathsf{T}\vec{y}$$

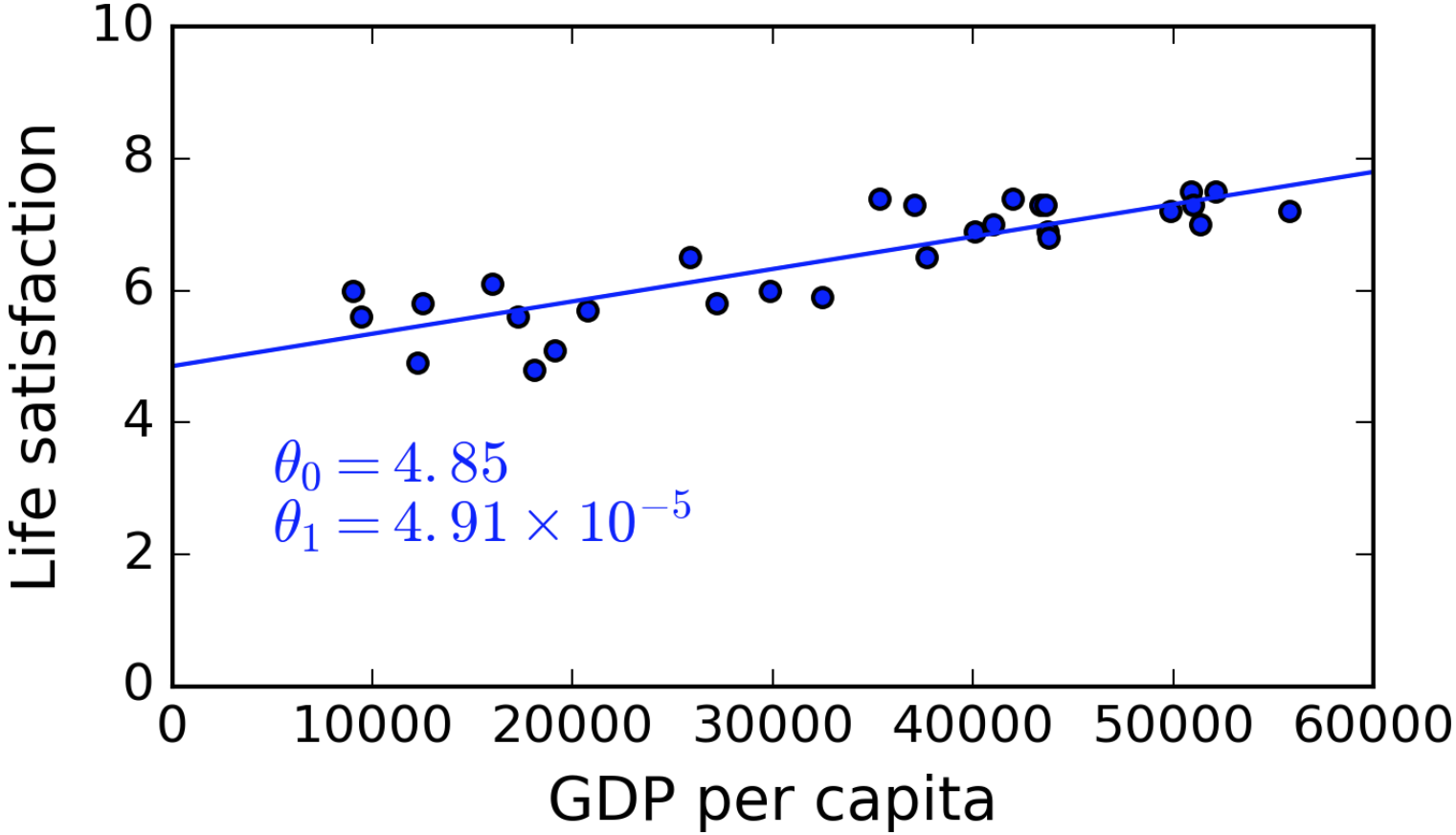
Optimial weights for ordinary linear regression

$$\hat{\vec{\theta}} = \begin{pmatrix} \hat{w}_0 \\ \hat{w}_1 \end{pmatrix} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\vec{y}$$

Adapted from material made by Andr

Fitted ordinary linear regression model

$$\texttt{life_satisfaction} = \hat{w}_0 + \hat{w}_1 \times \texttt{GDP_per_capita}$$
$$\texttt{life_satisfaction} = \hat{\theta}_0 + \hat{\theta}_1 \times \texttt{GDP_per_capita}$$



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Main challenges of machine learning

Irrelevant features

A critical part of the success of a Machine Learning project is coming up with a good set of features to train on: Feature engineering.

Poor quality data

If the training data is full of errors, outliers, and noise, it will make it harder for the system to detect the underlying patterns.

Overfitting

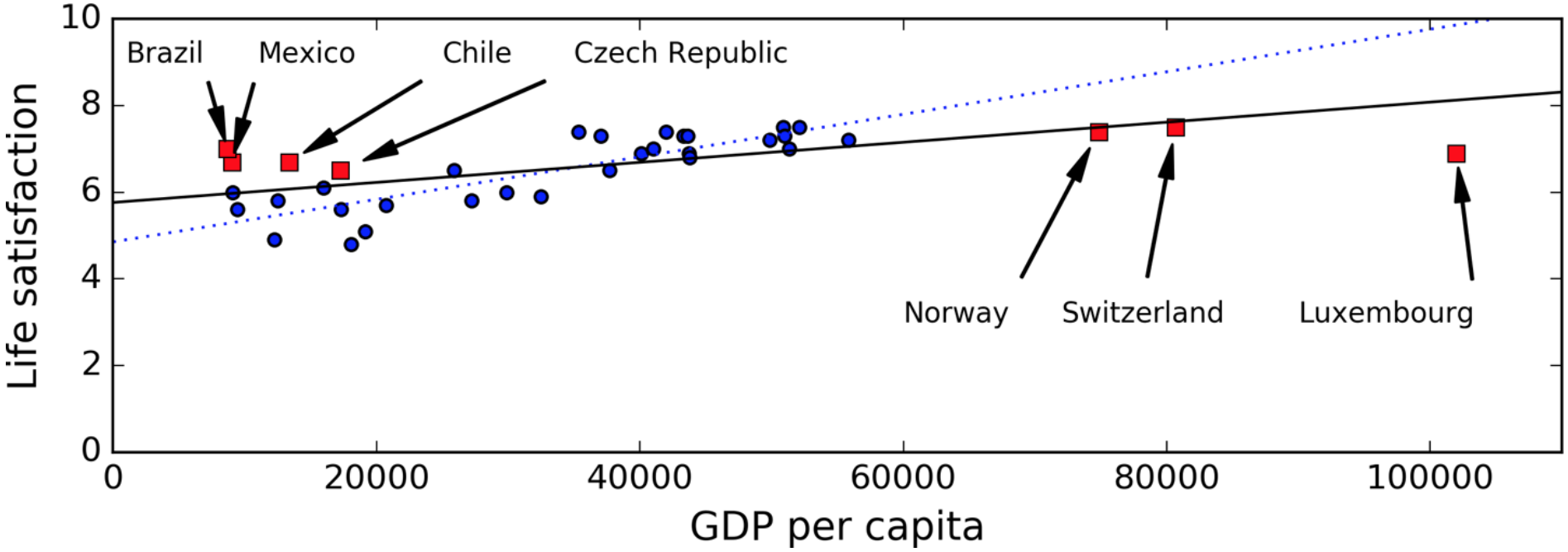
Overfitting happens when the model is too complex relative to the amount and noisiness of the training data.

Underfitting the data

Underfitting is the opposite of overfitting: it occurs when your model is too simple to learn the underlying structure of the data.

Nonrepresentative Training Data

In order to generalize well, it is crucial that your training data be representative of the new cases you want to generalize to.



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Poor-quality data

If the training data is full of errors, outliers, and noise, it will make it harder for the system to detect the underlying patterns.
If some instances are clearly outliers, it may help to simply discard them or try to fix the errors manually.
If some instances are missing a few features, you must decide

- whether you want to ignore this attribute (column of feature matrix) altogether,
- ignore these instances (row of feature matrix),
- fill in the missing values (imputing), or
- train one model with the feature and one model without it.

Irrelevant features

Garbage in, garbage out. A critical part of the success of a Machine Learning project is coming up with a good set of features to train on. This process, called *feature engineering*, involves:

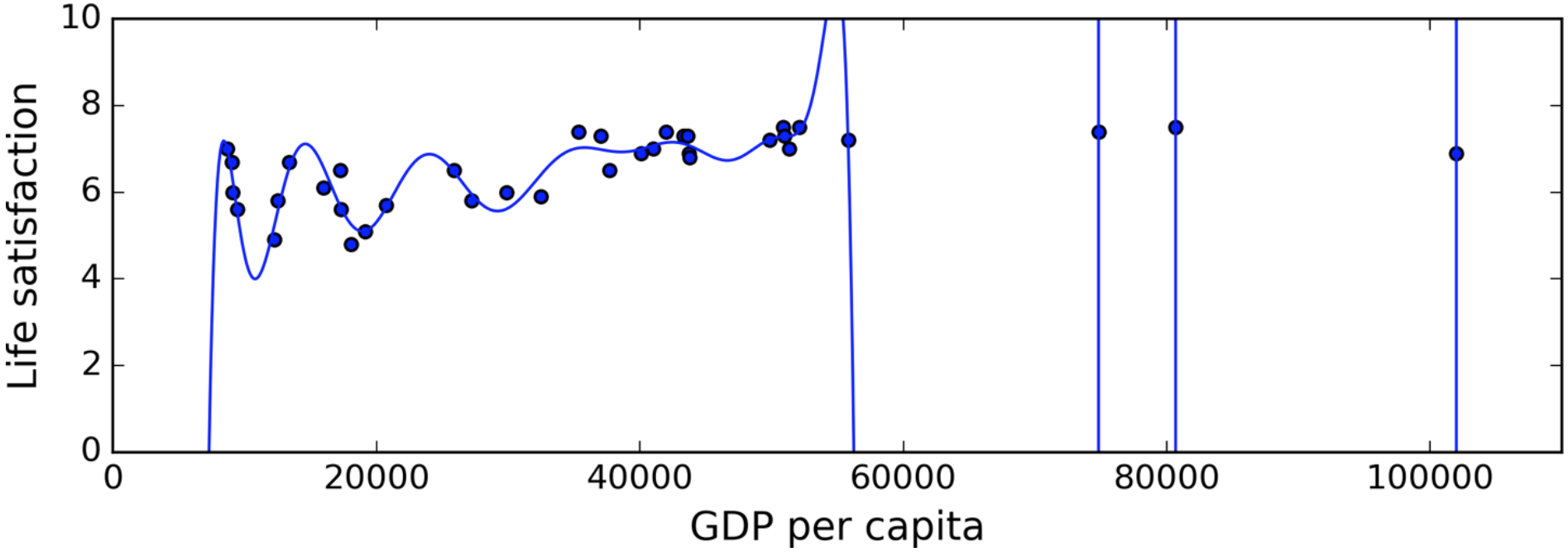
Feature selection selecting the most useful features to train on among existing features.

Feature extraction combining existing features to produce a more useful one.

Gathering new data.

Overfitting the training data

Overfitting happens when the model is too complex relative to the amount and noisiness of the training data.



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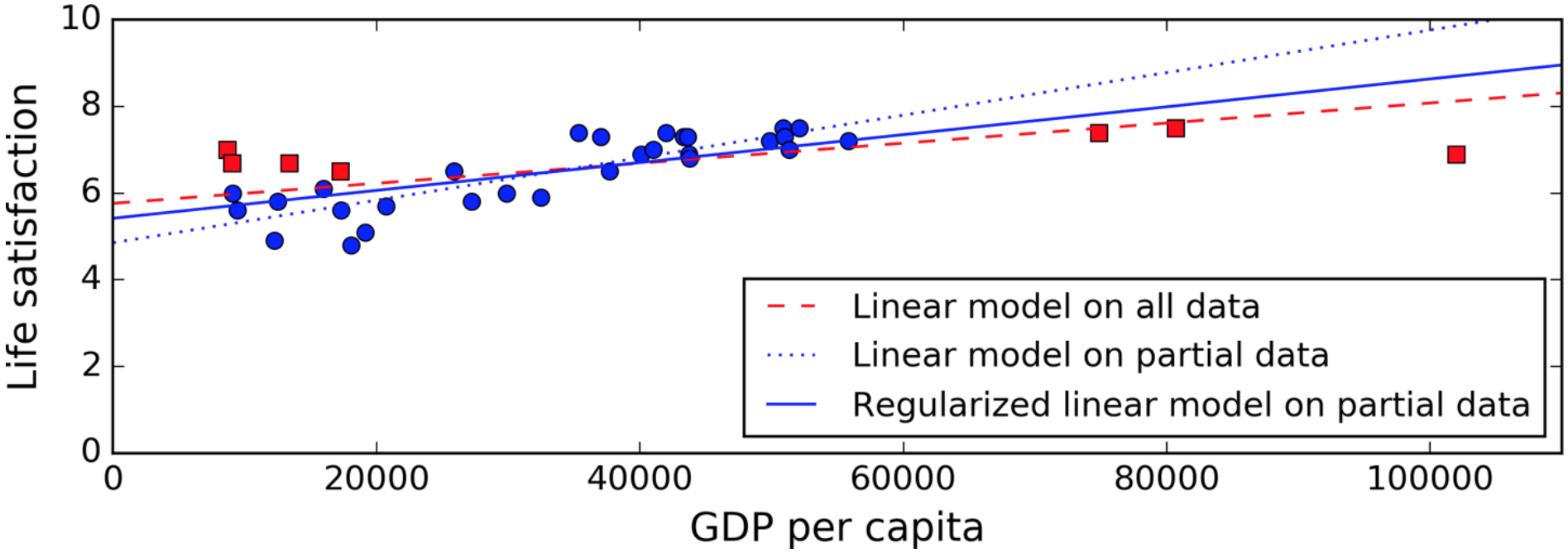
Circumvent the overfitting the training data:

- Simplify the model
- Gather more training data
- Reduce the noise in the training data
- Regularize the model



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Underfitting the data

Underfitting is the opposite of overfitting: it occurs when your model is too simple to learn the underlying structure of the data.
The main options to fix this problem are:

- Selecting a more powerful model, with more parameters
- Feeding better features to the learning algorithm (feature engineering)
- Reducing the constraints on the model (e.g., reducing the regularization hyperparameter)